

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of Bachelor of Technology (B.Tech)

University Examination November 2010

CE202 Data Communication and Networking

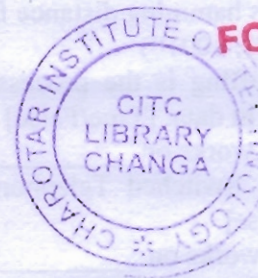
Date: 27.11.10, Saturday

Time: 10.30 a.m to 1.30 p.m

Maximum Marks: 70

Instructions:

1. Attempt all questions.
2. Answers both sections in separate answer books.
3. Figure to the right indicates full marks.
4. Assume suitable data, if necessary and mention it.



FOR REFERENCE

SECTION – I

Q.1 Answer the following questions.

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1. Bits from physical layer are converted into frames by the _____ layer.
2. Network layer of the OSI reference model provides end-to-end connection. [True/False]
3. HTTP works at the Application layer of the TCP/IP model. [True/False]
4. _____ is a device that transmits or receives data in the form of an analog or digital signal through a network.
5. Which address is used to locate application running on remote machine?
6. Define Baud Rate.
7. _____ is connection oriented protocol at transport layer.
8. _____ is the process in which the receiver tries to guess the message by redundancy bits.
9. FDM is a digital multiplexing technique. [True/False]
10. Define Composite Signals.
11. The period of a signal is 1000 ms. Calculate the frequency in kHz.

Q.2

- (a) Explain OSI reference model. Which layers of OSI are not available in TCP/IP model? 8
- (b) Write down the advantages and disadvantages of Mesh and Star topology. 4

OR

Q.2

- (a) What is protocol? Explain key elements of a protocol. What are the two categories of data communication standards? 6
- (b) Define the following addresses in TCP/IP. 6
 1. Physical Address
 2. Logical Address
 3. Port Address

Q.3

- (a) Describes various categories of network like LAN, MAN and WAN with example. 6
- (b) Explain Time Division Multiplexing. 6

OR

Q.3

- (a) What is guided transmission media? Write down the comparison of coaxial cable and twisted-pair. 6
- (b) Why does redundant bits require in Data Link Layer frame header? Compare LRC vs. VRC techniques. 6

SECTION – II

Q.4 Do as directed.

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1. Which type of error can not be detected by checksum?
2. Define Piggybacking.
3. Stop and Wait is a special case of Go Back N with window size 1. [True/False]
4. To create mesh topology with 5 nodes for simplex transmission media, total no of cables required is _____.
5. What is hamming distance for following codeword?
d(10101, 10010)
6. Bit stuffing is the process of adding 1 extra byte whenever there is an escape character in data. [True/False]
7. Bridge works on _____ layer.
8. Public Switched Telephone Network (PSTN) uses packet switching technique. [True/False]
9. HTTP works on _____ port number.
10. What is an RARP?
11. IP is connection oriented protocol. [True/False]

Q.5

- (a) Write down the comparisons of
1. Analog and Digital signals 2. Periodic and Aperiodic signals 6
- (b) What is transmission impairment? Explain the causes of transmission impairment. 6

OR

Q.5

- (a) Explain following terms with example:
1. Amplitude 2. Frequency 3. Phase 6
- (b) Explain Amplitude modulation with necessary diagram. 6

Q.6 Answer the following questions (Any TWO).

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- (a) Given a 10-bit sequence 1010011110 and a divisor of 1011, find the codeword.
- (b) Explain the various classes of Internet Protocol addresses.
- (c) Write a short note on Cable Modem.



FOR REFERENCE